Nisha Patel

(+1) 267 446 4419 | npatel61@gmu.edu | LinkedIn | GitHub

Technical Skills

Languages: Java, Python, JavaScript, C, C++
Backend: Spring Boot, Node.js, Express, REST APIs, Kafka, WebSockets
Frontend: React, Redux, JavaScript, Tailwind CSS, HTML/CSS
Cloud & DevOps: AWS (EC2, S3), Microsoft Azure, Docker, Kubernetes, CI/CD, Vercel
Databases: PostgreSQL, MySQL, MongoDB, MongoDB Atlas
Concepts: OOP, Data Structures & Algorithms, System Design, SDLC, Agile/Scrum, Microservices, Unit Testing
Tools: Git, GitHub, Postman, Linux, OpenCV

Professional Experience

Software Engineer | *Spring Boot, React, Kafka, Kubernetes* May 2024 – May 2025
Ekaantik Software Solutions

- Led full-stack development of a production digital communication platform using Spring Boot, React, Kafka, Docker, and Kubernetes supporting real-time messaging and notifications at scale.
- Architected WebSocket-based chat engine backed by MongoDB with message threading and read receipts achieving sub-100ms latency.
- Engineered RESTful API with JWT authentication, RBAC, and rate limiting enforcing multi-layer security controls across the user lifecycle.
- Managed Kubernetes manifests and Docker Compose configs enabling zero-downtime deployments, CI/CD pipelines, and horizontal pod autoscaling.

Software Engineering Intern | *React, Node.js, MongoDB, OpenCV* Jan 2023 – Jun 2023
Stat Modeller

- Built FaceSense, an AI-powered Visitor Recognition System using React, Node.js/Express, and Vercel delivering real-time facial recognition-based access control for enterprise clients.
- Developed Node.js REST API with paginated endpoints, event-driven logging, and real-time analytics processing hundreds of records per session at sub-second response times.
- Designed MongoDB Atlas cluster with optimized indexing, compound queries, and TTL policies for efficient multi-tenant data storage.
- Engineered OpenCV CCTV stream pipeline for live face detection and automated access event triggering from IP camera feeds.

Projects

Course RAG Tutor | *Next.js, TypeScript, Postgres/pgvector, Gemini, Vercel*

- Built an end-to-end RAG Q&A system over course PDFs with paragraph-aware chunking, Gemini embeddings, HNSW indexing, streaming cited answers, and multi-turn memory via LLM query rewriting.
- Implemented hybrid retrieval that merges pgvector cosine similarity with Postgres full-text search through Reciprocal Rank Fusion, so exact-term queries are matched alongside semantic ones.
- Built a 25-question evaluation harness (Precision@K, Recall@K, MRR) reaching an MRR of 0.888, and used it to trace failures to chunk dilution, a date-format bug in the eval set, and duplicate chunks fixed with idempotent ingestion.

Sandbox Escape Finder | *Python, AST, RestrictedPython, pytest*

- Built a tool that tests RestrictedPython against known Python sandbox-escape techniques by pairing five AST-based static detectors in a single shared tree walk with a dynamic payload prober.
- Designed an escape oracle that confirms an escape from any of three independent signals: a canary file outside the sandbox, a per-payload secret, and denylisted module loads.
- Against RestrictedPython 8.5, flagged all 10 attack payloads with zero escapes and zero false negatives, traced two to CVE-2023-37271 and CVE-2023-41039, and validated with a 17-test reproducible suite.

Interview AI | *React, Node.js, MongoDB, Gemini API*

- Built a full-stack AI interview prep platform with resume parsing, behavioral coaching, and job tracking, plus a voice-based AI interviewer using the Web Speech API for spoken Q&A and Gemini feedback.
- Added a salary negotiation simulator with AI recruiter roleplay, and in-browser coding via Monaco Editor and Judge0 with Gemini-generated DSA problems across 3 difficulty levels.

Education

George Mason University, Fairfax, Virginia Aug 2025 – May 2027
Master's in Software Engineering *GPA: 3.67*

CVM University Aug 2020 – May 2024
Bachelor of Computer Engineering *GPA: 3.4*